Notes on Quantum Jump Algorithm and SSE

Collection of notes, formulas and demonstrations in the field of dynamics of open quantum systems, *Stochastic Schrodinger Equation* and *Monte Carlo - Quantum Jump Algorithm*

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1 Stochastic Schrodinger Equation

Stochastic Schrodinger Equations are an alternative formulation of the evolution of an Open Quantum System, based on the time-dependent Schrodinger Equation, i.e. on the System *wave function*; so that they offer a less computationally expensive alternative for the evolution of the system over time, taking into account the effects of the environment, compared to the methods based on the *Density Matrix formalism.* In the Markovian limit a generic SSE reads :

$$i\frac{d}{dt}\left|\Psi_s(t)\right\rangle = \hat{H}_S(t)\left|\Psi_s(t)\right\rangle + \sum_q^M \alpha_q \hat{S}_q \left|\Psi_S(t)\right\rangle dW_t - \frac{i}{2}\sum_q^M \alpha_q^2 \hat{S}_q^\dagger \hat{S}_q \left|\Psi_S(t)\right\rangle \tag{1}$$

where α is the coupling strength, describing the intensity of the system-bath interaction; α^2 is the dissipation rate instead. dW_t represent the stochastic fluctuation, i.e. the noise of a Wiener process. Averaging over different Trajectories realization one can rebuild the *Density Matrix* as follows :

$$\rho_S(t) = \frac{1}{N_{Traj}} \sum_j^{N_{Traj}} \left|\Psi_{S,j}(t)\right\rangle \left\langle \Psi_{S,j}(t)\right| \tag{2}$$

where $|\Psi_{S,j}(t)\rangle$ is the system wave function corresponding to j-th realization, and expanding $|\Psi_{S,j}(t)\rangle$ to stationary eigenstates $|\phi_m\rangle$ of the system :

$$\left|\Psi_{S,j}(t)\right\rangle = \sum_m^{N_s tates} C_{m,j}(t)\left|\phi_m\right\rangle \tag{3}$$

from where it's possible to calculate the single term of the *Density Matrix* as :

$$\text{population} \quad q = \left(\rho_S(t)\right)_{qq} = \frac{1}{N_{Taj}} \sum_j^{N_{Traj}} \left|C_{q,j}(t)\right|^2 \tag{4}$$

$$\text{coherence} \quad k = \left(\rho_S(t)\right)_{qk} = \frac{1}{N_{Taj}} \sum_j^{N_{Traj}} C_{q,j}^*(t) C_{k,j}(t) \tag{5}$$

In the limit of large number of quantum trajectories, SSE reproduces the same $\rho_S(t)$ coming out from the Lindblad master equation.

The propagation of the *coefficients* $C(t)$ can be done with a second-order version of the Euler algorithm, leading to :

$$C(t+\delta t) = C(t-\delta t) - 2i\delta t H_{Eff} C(t) \tag{6}$$

1.1 Relaxation Fluctuation Theorem

Non Hermitian operators are fundamental to describe the loss of norm, typical of open quantum system, due to the fact that a single *wf* cannot include and represent the whole system probability. The Evolution Operator based on a Non-Hermitian Hamiltonian reads:

$$\hat{H}_{Eff} = \hat{H} - i\mathcal{D} \quad \text{so that} \quad \hat{H}_{Eff} \neq \hat{H}_{Eff}^\dagger \tag{7}$$

$$\text{where} \quad \mathcal{D} = \frac{\alpha^2}{2} \sum_q^M \hat{S}_q^\dagger \hat{S}_q \tag{8}$$

$$|\Psi(t+\delta t)\rangle = \mathcal{U} |\Psi(t)\rangle = e^{-i\hat{H}_{Eff}\delta t} |\Psi(t)\rangle = e^{-i\hat{H}\delta t - \mathcal{D}\Delta t} |\Psi(t)\rangle \tag{9}$$

where the hermitian part $e^{-i\hat{H}\delta t}$ give rise to a simple rotation, where the non - hermitian part $e^{-\mathcal{D}\delta t}$ gives an exponential decay instead, related to the loss of norm and so probability of the quantum state. In the optic of the **Relaxation Fluctuation Theorem**, the loss of norm induced by $\mathcal{D}$ gives rise to fluctuation on the System state.
SSE are based on this idea and indeed we can distinguish three specific term from Eq.(**??**) :

- $\hat{H}_S(t) |\Psi_s(t)\rangle$ represent the hermitian evolution (related to the deterministic force);
- $-\alpha^2 \frac{i}{2} \sum_q^M \hat{S}_q^\dagger \hat{S}_q |\Psi_S(t)\rangle$ represent the non hermitian evolution, interpretable as the dissipation due to the environment, that reduces the norm of the system vector (related to the deterministic damping force);
- $\alpha \sum_q^M W_q(t) \hat{S}_q |\Psi_S(t)\rangle$ represent the fluctuation induced by the dissipation, modeled as a Wiener process $dW_q(t)$ (or white noise $l_q(t)$,i.e. a noise associated with the Markov approximation (related to the stochastic force induced by the environment dissipation)

1.2 Generic form of SSE

It's important to point out that the SSE are restricted only to *Diffusive processes*, where the System is continuously modified by the environment effect, i.e. by a continuous Wiener noise; this corresponds to a *Continuously Monitored System.*

There is a general formulation to include quantum jump processes as well, which is called *Quantum Trajectories* or *Stochastic Unraveling* where the Wiener processes (dW) are accompanied by the *Poisson processes* (dN); the equation reads:

$$\begin{aligned} d|\tilde{\Psi}(t)\rangle = &\left[-i\hat{H} - \frac{1}{2}\sum_k \hat{S}_k^\dagger \hat{S}_k - \frac{1}{2}\sum_j \hat{L}_j^\dagger \hat{L}_j \right] |\tilde{\Psi}(t)\rangle\, dt \\ &+ \sum_k \hat{S}_k |\tilde{\Psi}(t)\rangle\, dW_k(t) \\ &+ \sum_j \left(\hat{L}_j - \hat{I} \right) |\tilde{\Psi}(t)\rangle\, dN_j(t) \end{aligned} \tag{10}$$

Using the *Ito Corrections* to introduce the renormalization directly in the equation of the evolution one obtains :

$$
\begin{aligned}
d\,|\Psi(t)\rangle = \Bigg[&- i\hat{H} - \frac{1}{2}\sum_k \left(\hat{S}_k^\dagger \hat{S}_k - 2\langle \hat{S}_k^\dagger \rangle_t \hat{S}_k + |\langle \hat{S}_k \rangle_t|^2 \right) \\
&- \frac{1}{2}\sum_j \left(\hat{L}_j^\dagger \hat{L}_j - \langle \hat{L}_j^\dagger \hat{L}_j \rangle_t \right) \Bigg] |\Psi(t)\rangle\, dt \\
&+ \sum_k \left(\hat{S}_k - \langle \hat{S}_k \rangle_t \right) |\Psi(t)\rangle\, dW_k(t) \\
&+ \sum_j \left(\frac{\hat{L}_j}{\sqrt{\langle \hat{L}_j^\dagger \hat{L}_j \rangle_t}} - \hat{I} \right) |\Psi(t)\rangle\, dN_j(t)
\end{aligned}
\tag{11}
$$

Where the mathematical terms are defined as follows:

- $|\Psi(t)\rangle$ and $|\tilde{\Psi}(t)\rangle$ represent the normalized and unnormalized quantum state vectors, respectively.
- $\hat{H}$ is the system Hamiltonian, a Hermitian operator representing the internal energy.
- $\hat{S}_k$ are the Lindblad operators for continuous measurement channels (diffusion processes).
- $\hat{L}_j$ are the Lindblad operators for discrete measurement channels (quantum jumps or Poisson processes).
- $\langle \hat{O} \rangle_t = \langle \Psi(t)|\, \hat{O}\, |\Psi(t)\rangle$ denotes the quantum expectation value of a generic operator $\hat{O}$ at time t.
- $dW_k(t)$ are independent Wiener increments representing continuous white noise, satisfying the Itô rule $dW_k^2 = dt$.
- $dN_j(t)$ are independent Poisson increments representing discrete random jumps, satisfying $dN_j^2 = dN_j$ (taking values of 0 or 1).
- $\hat{I}$ is the identity operator.

Remark : Trajectories obtained by the collision model algorithm are qualitatively different. In this case, the stochastic nature of the evolution can be more easily understood as resulting from the quantum probabilities for the outcomes of measurements performed on the ancilla environment.

1.3 Wiener Process

The *Wiener Process*, also called standard Brownian motion, is the mathematical model used to describe the stochastic time evolution driven by pure random continuous noise.

The stochastic Wiener process has to satisfy 4 fundamental properties :

- $W(0) = 0$ well known starting point;
- Markovian process and evolution, i.e. loss of memory

- the random trajectories are continuous in time, but since they are so incredibly jagged and chaotic that they are not derivable at any point;

- Gaussian increment, i.e. the variation of the process at every time step, follows a Normal Distribution, with $\langle W(t+\delta t)\rangle = 0$ and $\sigma^2_{dW} = Var\{dW\} = dt$; so the average evolution is zero, while the uncertainty grows with dt.

(Brief connection with the *Central Limit Theorem* (CLT) : the average over a finite number of sampling obtained with a Wiener process is zero at every time step, but the statistic error of the convergence of the average depends on $\frac{\sqrt{dt}}{\sqrt{N}}$, as defined by the CLT. The only difference is that in the CLT the distribution of the mean tends to become Gaussian as N increases, starting from any distribution, while in the Wiener process since the stochastic increment is already rigorously Gaussian by definition, the error follows this proportion in a mathematically exact way right from the start, without having to wait for N to tend to infinity**)**

A Wiener process that describe a Quantum State evolution would be reflected in a random trajectory of the state vector; but since that $\sigma_W = dt$ there will be a non zero term that contributes to the norm of the *wf*, i.e. $\langle\Psi|\Psi\rangle \approx 1 - dW^2 = 1 - dt$. So the noise itself would produce a deterministic drift that would "inflate" or "deflate" the total probability of the system.

For this reason, it is essential to renormalize the *wf* after each time step of the evolution; another possibility is to include, preemptively, a correction for the effect of norm loss due to random noise. This solution turns into a introduction of a non linear dependence, of the evolution, from the System state at every time step.

In this way you get the ***Quantum State Diffusion*** **(QSD)** formulation of the generic SSE :

$$\begin{aligned} d\,|\Psi_S(t)\rangle = &\left[-i\hat{H}_S(t) - \frac{\alpha^2}{2}\sum_q^M \left(\hat{S}_q^\dagger\hat{S}_q - 2\langle\hat{S}_q^\dagger\rangle_t\hat{S}_q + |\langle\hat{S}_q\rangle_t|^2\right)\right]|\Psi_S(t)\rangle\,dt \\ &+\alpha\sum_q^M\left(\hat{S}_q - \langle\hat{S}_q\rangle_t\right)|\Psi_S(t)\rangle\,dW_q(t) \end{aligned} \tag{12}$$

where the expectation values are defined continuously for the current state: $\langle\hat{S}_q\rangle_t = \langle\Psi_S(t)|\,\hat{S}_q\,|\Psi_S(t)\rangle$ and the noise increment satisfies Ito's rule: $dW_q^2 = dt$

2 SSE for Quantum Jump Algorithm

This paragraph is based on the article [1] and [2].

The *Quantum Jump* algorithm is based on the sequential evolution via a Non-Hermitian Deterministic Hamiltonian, followed by a random Jump Process, treated via a Monte Carlo scheme. The Non-Hermitian evolution is performed with the following Hamiltonian (see Eq.7) :

$$\hat{H}_{nH}(t) = \hat{H}_S(t) - \frac{i}{2}\sum_q^M \hat{S}_q^\dagger\hat{S}_q \tag{13}$$

which, for sufficient small δt gives :

$$|\Psi'_S(t+\delta t)\rangle = \left(1 - \frac{i\hat{H}_{nH}(t)\delta t}{\hbar}\right)|\Psi_S(t)\rangle \tag{14}$$

Since H is not Hermitian, this new wave function clearly is not normalized. The square of its norm is:

$$\begin{aligned}\langle\Psi'_S(t+\delta t)|\Psi'_S(t+\delta t)\rangle &= \langle\Psi_S(t)|\left(1 + \frac{i\hat{H}_{nH}(t)\delta t}{\hbar}\right)\left(1 - \frac{i\hat{H}_{nH}(t)\delta t}{\hbar}\right)|\Psi_S(t)\rangle \\ &= 1 - \delta p \end{aligned}\tag{15}$$

where δp reads :

$$\delta p = i\delta t\,\langle\Psi_S(t)|\left(\hat{H}_{nH} - \hat{H}^{\dagger}_{nH}\right)|\Psi_S(t)\rangle = \sum_q^M \delta p_q \tag{16}$$

$$\delta p_q = \delta t\,\langle\Psi_S(t)|\,S_q^{\dagger}S_q\,|\Psi_S(t)\rangle \tag{17}$$

The quantity δp represents the probability that a generic quantum jump occurs, while δp_q defines the probability that the quantum jump involves the specific interaction channel q, of relaxation or dephasing type.
It's important to notice that the norm loss δp is proportional to the time step δt, which by definition is very small, so that $\delta p << 1$. The second part of the evolution consist in a possible *quantum jump*, define as the sudden change of the wave function, that can correspond to the projection of the latter associated with a *gedanken measurement process*.
A *Monte Carlo Scheme* is used in order to decide wether the jump has occurred or not; in concrete terms, a quasi - random number ϵ,uniformly distributed between 0 and 1, is drawn and compared to δp :

- if $\epsilon > \delta p$, which happens most of the times (since $\delta p << 1$), no quantum jump occurs and the System *wf* evolves via only the Non-Hermitian Hamiltonian, so that the state at $t + \delta t$ reads :

$$|\Psi_S(t+\delta t)\rangle = \frac{|\Psi'_S(t+\delta t)\rangle}{(1-\delta p)^{1/2}} \tag{18}$$

- if $\epsilon \leq \delta p$ a quantum jump occurs, so that a new normalized state, defined by $S_q\,|\Psi_S(t)\rangle$, is chosen according to the probability $\Pi_q = {}^{\delta p_q}/_{\delta p}$;
  the System's state after the quantum jump is :

$$|\Psi_S(t+\delta t)\rangle = \frac{S_q\,|\Psi_S(t)\rangle}{\left({}^{\delta p_q}/_{\delta t}\right)^{1/2}} \tag{19}$$

$$\text{with probability} \quad \Pi_q = \frac{\delta p_q}{\delta p} \tag{20}$$

2.1 Equivalence with the Master Equation

The definition seen above can be used to build up a propagation algorithm of the System's wave function, that create different trajectories of evolution of the initial quantum states.
By definition the trajectories are random, so to obtain the effective Density Matrix at a certain time t, one has to average over all the different realization :

$$\overline{\rho(t)} = \overline{|\Psi_S(t)\rangle \langle \Psi_S(t)|} \tag{21}$$

It can be shown that

$$\overline{\rho(t)} = \rho_{QME}(t) \tag{22}$$

where $\rho_{QME}(t)$ represents the density matrix obtain via a *Quantum Master Equation* like the *Lindblad QME.*
Consider a Monte Carlo wave function (MCWF) at a certain time t $|\Psi_S(t)\rangle$; at the time $t + \delta t$ the average density matrix obtain for different random evolution reads:

$$\begin{aligned} \overline{\rho(t+\delta t)} = (1-\delta p) \frac{|\Psi'_S(t+\delta t)\rangle}{(1-\delta p)^{1/2}} \frac{\langle \Psi'_S(t+\delta t)|}{(1-\delta p)^{1/2}} \\ + \delta p \sum_q \Pi_q \frac{S_q |\Psi_S(t)\rangle}{(\delta p_q/\delta t)^{1/2}} \frac{\langle \Psi_S(t)| S_q^\dagger}{(\delta p_q/\delta t)^{1/2}} \end{aligned} \tag{23}$$

introducing Eq.14 one obtains :

$$\overline{\rho(t+\delta t)} = \rho(t) + i\delta t \left[\rho(t), \hat{H}_S\right] + \delta t \mathcal{D}\left[\rho(t)\right] \tag{24}$$

which in its differential form corresponds to the *Lindblad QME* :

$$\frac{d\overline{\rho(t)}}{dt} = i\left[\overline{\rho(t)}, \hat{H}_S\right] + \mathcal{D}\left[\overline{\rho(t)}\right] \tag{25}$$

2.2 Probability of a Quantum Jump

Considering a generic two level system with wave function :

$$|\Psi_S(t)\rangle = \alpha(t)|0\rangle + \beta(t)|1\rangle \tag{26}$$

where the dynamic describes the relaxation from the excited state $|1\rangle$ to the ground $|0\rangle$ with the operator :

$$\mathcal{D}(\rho_S) = -\frac{\Gamma}{2}\left(\sigma^+\sigma^-\rho_S + \rho_S\sigma^+\sigma^-\right) + \Gamma\sigma^-\rho_S\sigma^+ \tag{27}$$

The probability of performing a *Quantum Jump* that induce relaxation, from the excited state $|1\rangle$ to the ground $|0\rangle$ is proportional to the excited state's population :

$$\delta p = \Gamma|\beta|^2\delta t \tag{28}$$

where Γ is the relaxation rate, i.e. the spontaneous emission rate, corresponding to the inverse of the excited state lifetime.

If no *quantum jump* occurs, the evolution of the system is given by the Non-Hermitian Hamiltonian :

$$|\Psi_S(t+\delta t)\rangle = \alpha\left(1+\frac{\Gamma\delta t}{2}|\beta|^2\right)|0\rangle + \beta\left(1-\frac{\Gamma\delta t}{2}|\alpha|^2\right)\exp\left(-i\omega_S\delta t\right)|1\rangle \tag{29}$$

where ω_S represents the evolution associated to the System frequency [I].
It's important to notice that the intrinsic relaxation directly modifies the ground and excited populations, via a slight rotation of the wave function (i.e. the vector in the Bloch Sphere); this reduces the $|1\rangle$ population and so gives a lower Jump Probability after every time step where the jump has not occurred. So The probability $P(t)$ for having no quantum jumps between time 0 and t is found to be :

$$P(t) = |\alpha|^2 + |\beta|^2\exp\left(-\Gamma t\right) \tag{30}$$

where $|\alpha|^2$ is the probability that the state was already in $|0\rangle$, while $|\beta|^2\exp\left(-\Gamma t\right)$ represents the probability of being in $|1\rangle$ times the probability that the excitation survives, i.e. sum of the probabilities of the only two scenarios in which there is no relaxation.

Note I: *MCWF Physical Interpretation.*
In the Monte Carlo wave-function (MCWF) formalism, the dissipative evolution of a system is understood through a continuous measurement perspective. It consists of two complementary mechanisms:

- **Quantum Jump (Photon Detection):** This simulates the exact moment a photon is emitted and detected by the environment. The detection abruptly projects the wave function into the ground state.
- **Null Measurement (Continuous Evolution):** The absence of a detected photon (no jump) constitutes an active modification of the system's state. During this period, the system evolves under a non-Hermitian Hamiltonian. This continuous evolution smoothly rotates the state vector toward the ground state, decreasing the probability amplitude of the excited state. Physically, this reflects the increasing statistical certainty that, since no photon has been emitted over time, the system is less likely to jump in the future.

NOTE II : *Quantum Stochastic Master Equation*
To formulate a Stochastic Schrödinger Equation (SSE), projective measurements must be performed on the environment, which must be initialized in a pure state. This ensures that the principal system remains pure, implying that maximal information regarding the evolution process is continuously acquired. If these conditions are not satisfied, Stochastic Master Equations (SMEs) must be employed. SMEs operate directly on density matrices, thus accommodating the lack of complete information about the system, and their ensemble average rigorously recovers the deterministic Lindblad master equation.

3 Coccia System

We want to simulate a 3 level system with ground state $|0\rangle$ and the two excited states $|1\rangle$ and $|2\rangle$ with $\varepsilon_2 > \varepsilon_1$. The ground state is perturbatively excited into the state $|2\rangle$ so that the initial wave function reads:

$$|\Psi_S(0)\rangle = \sqrt{p_0}\,|0\rangle + \sqrt{p_2}\,|2\rangle \tag{31}$$

3.1 Stochastic Schrodinger Equation

Following the definition in [2] and [1] and setting $\hbar = 1$ we define the *Non - Hermitian Hamiltonian* as :

$$H_{nH} = H_S - \frac{i}{2}\sum_m C_m^\dagger C_m \tag{32}$$

with :

- pure dephasing $\rightarrow \quad C_d = \sqrt{\frac{\Gamma_d}{2}}\left(|2\rangle\langle 2| - |1\rangle\langle 1|\right) = \sqrt{\frac{\Gamma_d}{2}}\sigma_z^{21} \quad \rightarrow \quad C_d^\dagger C_d = \frac{\Gamma_d}{2}\left(|1\rangle\langle 1| + |2\rangle\langle 2|\right)$
- 2-1 relaxation $\rightarrow \quad C_r = \sqrt{\Gamma_r}\,|1\rangle\langle 2| = \sqrt{\Gamma_r}\sigma_-^{21} \quad \rightarrow \quad C_r^\dagger C_r = \Gamma_r\,|2\rangle\langle 2|$

With a first order approximation we can define the Evolution Operator as :

$$\mathcal{U}(\Delta t) \approx \mathbb{I} - iH_{nH}\Delta t + \mathcal{O}\left(\Delta t^2\right) \tag{33}$$

so that the evolved System's wave function reads :

$$\begin{aligned}
|\Psi_S'(\Delta t)\rangle &= \mathcal{U}(\Delta t)\,|\Psi_S(0)\rangle = \left(\mathbb{I} - iH_{nH}\Delta t\right)|\Psi_S(0)\rangle \\
&= |\Psi_S(0)\rangle - i\Delta t\left[H_S\,|\Psi_S(0)\rangle - \frac{i\Gamma_d}{4}\left(|1\rangle\langle 1| + |2\rangle\langle 2|\right)|\Psi_S(0)\rangle - \frac{i\Gamma_r}{2}\,|2\rangle\langle 2|\,|\Psi_S(0)\rangle\right] \\
&= |\Psi_S(0)\rangle - i\Delta t\left(\varepsilon_0\sqrt{p_0}\,|0\rangle + \varepsilon_2\sqrt{p_2}\,|2\rangle\right) - \frac{\Delta t\Gamma_d}{4}\sqrt{p_2}\,|2\rangle - \frac{\Delta t\Gamma_r}{2}\sqrt{p_2}\,|2\rangle \\
&= \sqrt{p_0}\left(1 - i\Delta t\varepsilon_0\right)|0\rangle + \sqrt{p_2}\left(1 - \frac{\Delta t}{4}\left(\Gamma_d + 2\Gamma_r\right) - i\Delta t\varepsilon_2\right)|2\rangle
\end{aligned} \tag{34}$$

No population in $|1\rangle$ but reduction of the population in $|2\rangle$. Since the evolution occurred with a Non - Hermitian Hamiltonian the state after the wave function is non normalized, so that we need to calculate the square of the norm as:

$$\begin{aligned}
\langle\Psi_S'(t+\Delta t)|\Psi_S'(t+\Delta t)\rangle &= \langle\Psi_S(0)|\left(\mathbb{I} + iH_{nH}^\dagger\Delta t\right)\left(\mathbb{I} - iH_{nH}\Delta t\right)|\Psi_S(0)\rangle \\
&= 1 - i\Delta t\,\langle\Psi_S(0)|\,H_{nH} - H_{nH}^\dagger\,|\Psi_S(0)\rangle \\
&= 1 - i\Delta t\,\langle\Psi_S(0)|\left(H_S - \frac{i}{2}\sum_m C_m^\dagger C_m\right) - \left(H_S + \frac{i}{2}\sum_m C_m^\dagger C_m\right)|\Psi_S(0)\rangle \\
&= 1 - \Delta t\,\langle\Psi_S(0)|\sum_m C_m^\dagger C_m\,|\Psi_S(0)\rangle = 1 - \delta p
\end{aligned} \tag{35}$$

making explicit δp we obtain :

$$\delta p = \Delta t\,\langle\Psi_S(0)|\sum_m C_m^\dagger C_m\,|\Psi_S(0)\rangle$$

$$
\begin{aligned}
&= \Delta t \left\langle \Psi_S(0) \right| \frac{\Gamma_d}{2} \left(\left|1\right\rangle \left\langle 1\right| + \left|2\right\rangle \left\langle 2\right| \right) + \Gamma_r \left|2\right\rangle \left\langle 2\right| \left|\Psi_S(0)\right\rangle \\
&= \Delta t \, p_2 \left(\frac{\Gamma_d}{2} + \Gamma_r \right)
\end{aligned} \tag{36}
$$

This result coincides with the complete calculation of the squared norm, neglecting the terms with Δt^2 (limit $\Delta t \to 0$) :

$$
\begin{aligned}
\langle \Psi'_S(t+\Delta t) | \Psi'_S(t+\Delta t) \rangle &= p_0 \left(1 + \Delta t^2 \varepsilon_0^2\right) + p_2 \left[\left(1 - \frac{\Delta t}{4} \left(\Gamma_d + 2\Gamma_r\right)\right)^2 + \Delta t^2 \varepsilon_2^2 \right] \\
&= p_0 + p_2 \left(1 - \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)\right) = p_0 + p_2 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right) \\
&= 1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right) = 1 - \delta p
\end{aligned} \tag{37}
$$

So now the normalized evolved state reads :

$$
\left|\Psi_S\left(\Delta t\right)\right\rangle = \frac{\sqrt{p_0}\left(1 - i\Delta t \varepsilon_0\right)}{\sqrt{1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)}} \left|0\right\rangle + \frac{\sqrt{p_2}\left(1 - \frac{\Delta t}{4}\left(\Gamma_d + 2\Gamma_r\right) - i\Delta t \varepsilon_2\right)}{\sqrt{1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)}} \left|2\right\rangle \tag{38}
$$

The state population now are :

$$
p_0\left(\Delta t\right) = \left\langle \Psi_S | 0 \right\rangle \left\langle 0 | \Psi_S \right\rangle = \left(\frac{\sqrt{p_0}\left(1 - i\Delta t \varepsilon_0\right)}{\sqrt{1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)}} \right)^2 = \frac{p_0\left(1 + \Delta t^2 \varepsilon_0^2\right)}{1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)} \approx \frac{p_0}{1 - \delta p} \tag{39}
$$

$$
p_1\left(\Delta t\right) = \left\langle \Psi_S | 1 \right\rangle \left\langle 1 | \Psi_S \right\rangle = 0 \tag{40}
$$

$$
\begin{aligned}
p_2\left(\Delta t\right) &= \left\langle \Psi_S | 2 \right\rangle \left\langle 2 | \Psi_S \right\rangle = \left(\frac{\sqrt{p_2}\left(1 - \frac{\Delta t}{4}\left(\Gamma_d + 2\Gamma_r\right) - i\Delta t \varepsilon_2\right)}{\sqrt{1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)}} \right)^2 \\
&= \frac{p_2 \left[\left(1 - \frac{\Delta t}{4}\left(\Gamma_d + 2\Gamma_r\right)\right)^2 + \Delta t^2 \varepsilon_2^2 \right]}{1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)} \\
&= \frac{p_2 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)}{1 - p_2 \Delta t \left(\frac{\Gamma_d}{2} + \Gamma_r\right)} \approx \frac{p_2 - \delta p}{1 - \delta p} = \frac{p_2 - \delta p}{p_0 + p_2 - \delta p}
\end{aligned} \tag{41}
$$

3.2 Collisional Methods

We now try to do the same unraveling but via *Collisional Methods.* Note that most of the Equation used are derived in the notes MC, where we treated a simple two level system; here the dynamic will be the same, i.e relaxation of an excited state, but extending the system to 3 levels, with $|0\rangle$ not involved in the collision. The initial System wave function is still :

$$|\Psi_S(0)\rangle = \sqrt{p_0}\,|0\rangle + \sqrt{p_2}\,|2\rangle \tag{42}$$

The Hamiltonian reads :

$$H_{CM} = H_S + H_{Coll} \tag{43}$$

with H_S associated to the eigenenergies of the system, while the Collisional Hamiltonian reads:

$$H_{Coll} = c\left(|1\rangle\langle 2| \otimes \sigma_+^a + |2\rangle\langle 1| \otimes \sigma_-^a\right) \tag{44}$$

We use an ancilla always initialize in $|0_a\rangle$ so that $\rho_a = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.
The total wave function is $|\Psi(0)\rangle = |\Psi_S(0)\rangle \otimes |0_a\rangle$.
The collisional step of evolution is carried out with :

$$\mathcal{U}_{Coll}\left(\Delta t\right) = \exp\left(-iH_{Coll}\Delta t\right) = |0\rangle\langle 0| \otimes \mathbb{I}^a + P + \cos\left(c\Delta t\right) Z - i\sin\left(c\Delta t\right)\frac{X}{2} \tag{45}$$

the derivation is the same of the notes MC but in this case from the Taylor expansion of the exponential the first term, i.e. Identity acts on the total system reading $\mathbb{I}^{tot} = (|0\rangle\langle 0| + |1\rangle\langle 1| + |2\rangle\langle 2|)\otimes \mathbb{I}^a$; the first term is the one appearing in $\mathcal{U}_{Coll}$, while the other two end up in the McLaurin series for the cos and sin.
The other terms read :

$$P = \frac{(|1\rangle\langle 1| + |2\rangle\langle 2|) \otimes \mathbb{I}^a + (|1\rangle\langle 1| - |2\rangle\langle 2|) \otimes \sigma_z^a}{2} \tag{46}$$

$$Z = \frac{(|1\rangle\langle 1| + |2\rangle\langle 2|) \otimes \mathbb{I}^a - (|1\rangle\langle 1| - |2\rangle\langle 2|) \otimes \sigma_z^a}{2} \tag{47}$$

$$X = (|1\rangle\langle 2| + |2\rangle\langle 1|) \otimes \sigma_x^a - i\,(|1\rangle\langle 2| - |2\rangle\langle 1|) \otimes \sigma_y^a \tag{48}$$

Now we can apply every single term to the total wave function :

$$\begin{aligned} P\left(|\Psi_S\rangle \otimes |0_a\rangle\right) &= \frac{(|1\rangle\langle 1| + |2\rangle\langle 2|) \otimes \mathbb{I}^a + (|1\rangle\langle 1| - |2\rangle\langle 2|) \otimes \sigma_z^a}{2}\left(\sqrt{p_0}\,|0\rangle + \sqrt{p_2}\,|2\rangle\right) \otimes |0_a\rangle \\ &= \frac{\sqrt{p_2}\,|2\rangle \otimes |0_a\rangle - \sqrt{p_2}\,|2\rangle \otimes |0_a\rangle}{2} = 0 \end{aligned} \tag{49}$$

$$\begin{aligned} Z\left(|\Psi_S\rangle \otimes |0_a\rangle\right) &= \frac{(|1\rangle\langle 1| + |2\rangle\langle 2|) \otimes \mathbb{I}^a - (|1\rangle\langle 1| - |2\rangle\langle 2|) \otimes \sigma_z^a}{2}\left(\sqrt{p_0}\,|0\rangle + \sqrt{p_2}\,|2\rangle\right) \otimes |0_a\rangle \\ &= \frac{\sqrt{p_2}\,|2\rangle \otimes |0_a\rangle + \sqrt{p_2}\,|2\rangle \otimes |0_a\rangle}{2} = \sqrt{p_2}\,|2\rangle \otimes |0_a\rangle \end{aligned} \tag{50}$$

$$\begin{aligned} X\left(|\Psi_S\rangle \otimes |0_a\rangle\right) &= \left[(|1\rangle\langle 2| + |2\rangle\langle 1|) \otimes \sigma_x^a - i\,(|1\rangle\langle 2| - |2\rangle\langle 1|) \otimes \sigma_y^a\right]\left(\sqrt{p_0}\,|0\rangle + \sqrt{p_2}\,|2\rangle\right) \otimes |0_a\rangle \\ &= \sqrt{p_2}\,|1\rangle \otimes |1_a\rangle + \sqrt{p_2}\,|1\rangle \otimes |1_a\rangle = 2\sqrt{p_2}\,|1\rangle \otimes |1_a\rangle \end{aligned} \tag{51}$$

The total evolved wave function, including the identity term for $|0\rangle$ now reads :

$$|\Psi(\Delta t)\rangle = \mathcal{U}_{Coll}|\Psi(0)\rangle = \sqrt{p_0}\,|0\rangle\otimes|0_a\rangle + \cos(c\Delta t)\sqrt{p_2}\,|2\rangle\otimes|0_a\rangle - i\sin(c\Delta t)\sqrt{p_2}\,|1\rangle\otimes|1_a\rangle \quad (52)$$

If we measure the Ancilla in state $|0_a\rangle$ the new System state is :

$$|\Psi_S(\Delta t)\rangle_0 = \sqrt{p_0}\,|0\rangle + \cos(c\Delta t)\sqrt{p_2}\,|2\rangle \quad (53)$$

this state is not normalized so we can calculate the squared norm as :

$$\begin{aligned}\langle\Psi_S(\Delta t)|\Psi_S(\Delta t)\rangle &= p_0 + p_2\cos^2(c\Delta t)\\ &= 1 - p_2\left(1-\cos^2(c\Delta t)\right)\\ &= 1 - p_2\sin^2(c\Delta t) = 1 - \delta p \end{aligned} \quad (54)$$

where δp is the probability of measuring the Ancilla in state $|1_a\rangle$, i.e. have a *Quantum Jump.*

The normalized wave function reads :

$$|\Psi_S(\Delta t)\rangle_0 = \frac{\sqrt{p_0}}{\sqrt{1-\delta p}}\,|0\rangle + \frac{\cos(c\Delta t)\sqrt{p_2}}{\sqrt{1-\delta p}}\,|2\rangle \quad (55)$$

The population of $|0\rangle$ and $|2\rangle$ are :

$$\langle\Psi_S(\Delta t)|0\rangle\langle 0|\Psi_S(\Delta t)\rangle = \frac{p_0}{1-\delta p} \quad (56)$$

$$\langle\Psi_S(\Delta t)|1\rangle\langle 1|\Psi_S(\Delta t)\rangle = 0 \quad (57)$$

$$\langle\Psi_S(\Delta t)|2\rangle\langle 2|\Psi_S(\Delta t)\rangle = \frac{\cos^2(c\Delta t)\,p_2}{1-\delta p} \quad (58)$$

3.3 Kraus Operators

Once we defined $\mathcal{U}_{Coll}$ we can define the associated *Kraus Operators*, as defined in the *Petruccione* :

$$\rho_S(t) = \mathcal{V}(t)\,\rho_S(0) = Tr_B\Big\{\mathcal{U}(t,t_0)\Big[\rho_S\otimes\rho_B\Big]\mathcal{U}^\dagger(t,t_0)\Big\} \quad (59)$$

$$\left(\rho_B = \sum_\alpha \lambda_\alpha|\varphi_\alpha\rangle\langle\varphi_\alpha| \quad \text{and} \quad Tr_B = \sum_\beta\langle\varphi_\beta|\ldots|\varphi_\beta\rangle\right)$$

$$\begin{aligned} &= \sum_\alpha\sum_\beta \lambda_\alpha\langle\varphi_\beta|\mathcal{U}(t,t_0)\left[\rho_S\otimes|\varphi_\alpha\rangle\langle\varphi_\alpha|\right]\mathcal{U}^\dagger(t,t_0)|\varphi_\beta\rangle\\ &= \sum_{\alpha,\beta}\mathcal{K}_{\alpha,\beta}(t)\,\rho_S\mathcal{K}^\dagger_{\alpha,\beta}(t) \end{aligned} \quad (60)$$

In our case, since we have $\rho_a = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}$, with only one non zero eigenvalue we obtain only two *Kraus Operators* in the two opposite limits :

Quantum Jump Limit Measurement base : $\{|0\rangle, |1\rangle\}$

$$\mathcal{K}_{0,0}^{QJ} = |0\rangle\langle 0| + |1\rangle\langle 1| + \cos(c\Delta t)|2\rangle\langle 2| \tag{61}$$

$$\mathcal{K}_{0,1}^{QJ} = -i\sin(c\Delta t)|1\rangle\langle 2| \tag{62}$$

We can check the Completeness Relation :

$$\begin{aligned}
\mathcal{K}_{0,0}^{QJ\dagger}\mathcal{K}_{0,0}^{QJ} &= |0\rangle\langle 0| + |1\rangle\langle 1| + \cos^2(c\Delta t)|2\rangle\langle 2| \\
\mathcal{K}_{0,1}^{QJ\dagger}\mathcal{K}_{0,1}^{QJ} &= \sin^2(c\Delta t)|2\rangle\langle 2| \\
\sum_{\alpha,\beta}\mathcal{K}_{\alpha,\beta}^{QJ\dagger}\mathcal{K}_{\alpha,\beta}^{QJ} &= \mathcal{K}_{0,0}^{\dagger}\mathcal{K}_{0,0} + \mathcal{K}_{0,1}^{\dagger}\mathcal{K}_{0,1} \\
&= |0\rangle\langle 0| + |1\rangle\langle 1| + \cos^2(c\Delta t)|2\rangle\langle 2| + \sin^2(c\Delta t)|2\rangle\langle 2| \\
&= |0\rangle\langle 0| + |1\rangle\langle 1| + |2\rangle\langle 2| = \mathbb{I}
\end{aligned} \tag{63}$$

The associated Probability are :

$$P_{0_a}^{QJ} = \langle\Psi_S|\mathcal{K}_{0,0}^{QJ\dagger}\mathcal{K}_{0,0}^{QJ}|\Psi_S\rangle = p_0 + p_2\cos^2(c\Delta t) \tag{64}$$

$$P_{1_a}^{QJ} = \langle\Psi_S|\mathcal{K}_{0,1}^{QJ\dagger}\mathcal{K}_{0,1}^{QJ}|\Psi_S\rangle = p_2\sin^2(c\Delta t) \tag{65}$$

State Diffusion Limit Measurement base : $\{|+\rangle, |-\rangle\}$ with $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$

$$\mathcal{K}_{+,0}^{SD} = \frac{|0\rangle\langle 0| + |1\rangle\langle 1| + \cos(c\Delta t)|2\rangle\langle 2| - i\sin(c\Delta t)|1\rangle\langle 2|}{\sqrt{2}} = \frac{\mathcal{K}_{0,0}^{QJ} + \mathcal{K}_{0,1}^{QJ}}{\sqrt{2}} \tag{66}$$

$$\mathcal{K}_{-,0}^{SD} = \frac{|0\rangle\langle 0| + |1\rangle\langle 1| + \cos(c\Delta t)|2\rangle\langle 2| + i\sin(c\Delta t)|1\rangle\langle 2|}{\sqrt{2}} = \frac{\mathcal{K}_{0,0}^{QJ} - \mathcal{K}_{0,1}^{QJ}}{\sqrt{2}} \tag{67}$$

The associated Probability are :

$$P_{+_a}^{SD} = \langle\Psi_S|\mathcal{K}_{0,0}^{SD\dagger}\mathcal{K}_{0,0}^{SD}|\Psi_S\rangle = \frac{1}{2}\left[p_0 + p_2\left(\cos^2(c\Delta t) + \sin^2(c\Delta t)\right)\right] = \frac{1}{2} \tag{68}$$

$$P_{-_a}^{SD} = \langle\Psi_S|\mathcal{K}_{0,1}^{SD\dagger}\mathcal{K}_{0,1}^{SD}|\Psi_S\rangle = \frac{1}{2}\left[p_0 + p_2\left(\cos^2(c\Delta t) + \sin^2(c\Delta t)\right)\right] = \frac{1}{2} \tag{69}$$

4 Fundamental Concepts of Stochastic Calculus

In the context of open quantum systems, stochastic calculus is essential for describing continuous measurements and unravelings such as State Diffusion and Quantum Jumps. The two main mathematical objects used to model the noise are the Wiener process and the Poisson process.

4.1 The Wiener Process

The Wiener process models continuous white noise, typical of Brownian motion and homodyne detection schemes. The infinitesimal increment dW represents the evolution of the noise over the time interval dt. Its fundamental statistical properties are defined by its expectation value and variance:

$$\mathbb{E}[dW] = 0, \quad \mathbb{E}[dW^2] = dt \tag{70}$$

Physically, this means that the noise fluctuates symmetrically around zero, but its amplitude scales with the square root of time, i.e., $dW \propto \sqrt{dt}$.

4.2 Itô Algebra

Because dW scales as $\sqrt{dt}$, second-order terms cannot be neglected when performing differential expansions, unlike in classical calculus where $dt^2 \to 0$. In Itô calculus, the following strict algebraic rules apply when manipulating stochastic differentials:

$$dW^2 = dt \tag{71}$$

$$dW dt = 0 \tag{72}$$

$$dt^2 = 0 \tag{73}$$

The relation $dW^2 = dt$ is the cornerstone of Itô calculus: the square of the noise increment becomes completely deterministic.

4.3 Itô's Lemma

Due to the non-negligible second-order terms, the standard Taylor expansion for a function of a stochastic variable is modified. If x is a variable governed by a stochastic differential equation, the differential of a function $f(x)$ must include the second derivative:

$$df(x) = f'(x)dx + \frac{1}{2}f''(x)(dx)^2 \tag{74}$$

When expanding $(dx)^2$, any $(dW)^2$ terms evaluate to dt and contribute to the deterministic drift of the equation. This specific second-order contribution is the origin of the non-linear compensation terms in the standard Quantum State Diffusion formalism.

4.4 The Poisson Process

For discontinuous measurement records, such as photon counting or quantum jumps, the noise is initially modeled using a macroscopic discrete Poisson process. Let $N(t)$ be the total number of events up to time t. Over a finite time interval Δt, assuming a mean event rate λ, the probability of observing exactly k events is governed by the Poisson distribution:

$$P(N(t) = k) = \frac{(\lambda \Delta t)^k e^{-\lambda \Delta t}}{k!} \tag{75}$$

A fundamental property of the macroscopic Poisson distribution is that both its expectation value and its variance are exactly equal to $\lambda \Delta t$:

$$\mathbb{E}[N(t)] = \sum_{k=0}^{\infty} k P(k) = \sum_{k=0}^{\infty} k \frac{(\lambda t)^k e^{-\lambda t}}{k!} = \lambda \Delta t \tag{76}$$

$$\mathrm{Var}(N(t)) = \mathbb{E}[N(t)^2] - (\mathbb{E}[N(t)])^2 = \lambda \Delta \tag{77}$$

In the context of stochastic differential equations, we focus on the infinitesimal increment $dn = N(t+dt) - N(t)$, which represents the number of jumps occurring in the microscopic time step dt. To understand the statistical behavior of dn, we take the limit $\Delta t \to dt$ and expand the exponential term to the first order, since terms of order dt^2 are negligible in Itô calculus:

$$e^{-\lambda dt} = 1 - \lambda dt + \mathcal{O}(dt^2) \approx 1 - \lambda dt \tag{78}$$

By substituting this expansion into the Poisson probability formula, we can evaluate the probabilities of observing a specific number of events k in the strictly infinitesimal interval dt:

$$P(0) = \frac{(\lambda dt)^0 e^{-\lambda dt}}{0!} \approx 1 - \lambda dt \tag{79}$$

$$P(1) = \frac{(\lambda dt)^1 e^{-\lambda dt}}{1!} = \lambda dt(1 - \lambda dt) \approx \lambda dt \tag{80}$$

$$P(k \geq 2) \propto \mathcal{O}(dt^2) \approx 0 \tag{81}$$

Because the probability of two or more simultaneous jumps strictly vanishes ($dt^2 = 0$), the sample space of the variable dn collapses. The increment can only assume the binary values 0 (no jump) or 1 (one jump). Consequently, in the infinitesimal limit, the Poisson distribution asymptotically reduces to a Bernoulli distribution.

We can now analytically compute the expectation value and variance for this infinitesimal binary variable. The expected value is simply determined by the probability of a jump occurring:

$$\mathbb{E}[dn] = 0 \cdot P(0) + 1 \cdot P(1) = \lambda dt \tag{82}$$

The variance is defined as $\mathrm{Var}(dn) = \mathbb{E}[dn^2] - (\mathbb{E}[dn])^2$. Since dn exclusively takes values of 0 or 1, it acts as an idempotent variable, meaning $dn^2 = dn$. Therefore, its second moment is equivalent to its first moment: $\mathbb{E}[dn^2] = \lambda dt$. Substituting this into the variance formula yields:

$$\mathrm{Var}(dn) = \lambda dt - (\lambda dt)^2 = \lambda dt \tag{83}$$

The $(\lambda dt)^2$ term completely disappears because $dt^2 = 0$. Remarkably, the fundamental Poisson property where the mean equals the variance is strictly preserved even in the infinitesimal Bernoulli limit. Note that another characteristic of the Bernoulli process emerge here, i.e. that the variance is the product of the two distinct probabilities:

$$\mathrm{Var}(dn) = P_0 \, P_1 = \lambda dt \, (1 - \lambda dt) = \lambda dt - (\lambda dt)^2 = \lambda dt \tag{84}$$

Finally, this idempotent nature mathematically justifies the algebraic substitution rules for the Poisson increment used to expand stochastic differentials:

$$dn^2 = dn \tag{85}$$

$$dndt = 0 \tag{86}$$

5 Generic Forms of Stochastic Schrödinger Equations

In the theory of open quantum systems, the continuous measurement of the environment leads to stochastic trajectories for the pure state of the system. Depending on the measurement scheme adopted (the unraveling), the Stochastic Schrödinger Equation (SSE) takes different forms. The two most prominent unravelings are the discrete Quantum Jumps (associated with direct detection) and the continuous Quantum State Diffusion (associated with homodyne detection). Assuming a single decay channel with rate Γ and a generic jump operator c, the explicitly non-linear standard forms are given below.

5.1 Quantum Jump (QJ) Unraveling

The Quantum Jump SSE describes a system undergoing a smooth, non-unitary deterministic evolution interrupted by abrupt, discrete jumps. The stochastic variable is the Poisson increment dn, which equals 1 with probability $P = \Gamma\langle c^\dagger c\rangle dt$, and 0 otherwise. The generic non-linear equation reads:

$$d|\psi(t)\rangle = -\frac{i}{\hbar}H|\psi(t)\rangle\,dt - \frac{1}{2}\Gamma\left(c^\dagger c - \langle c^\dagger c\rangle\right)|\psi(t)\rangle\,dt + \left(\frac{c}{\sqrt{\langle c^\dagger c\rangle}} - \mathbb{I}\right)|\psi(t)\rangle\,dn \tag{87}$$

The term proportional to dt represents the deterministic no-jump evolution, which continuously preserves the norm of the state vector. The term proportional to dn acts only when a measurement click is registered, instantaneously projecting the system into the new normalized state.

5.2 Quantum State Diffusion (SD) Unraveling

The Quantum State Diffusion (SD) SSE describes an unraveling driven by continuous Gaussian noise. However, depending on the specific physical measurement scheme (e.g., measuring one or two conjugate quadratures of the environment), the equation takes two distinct standard forms.

Homodyne Detection (Real Noise Unraveling)

If the measurement apparatus extracts a single continuous stream of real classical information (homodyne detection), the system is driven by a real Wiener increment dW, which satisfies the Itô relation $dW^2 = dt$. The explicitly norm-preserving, non-linear equation requires compensating only for the Hermitian part of the jump operator:

$$d|\psi(t)\rangle = -\frac{i}{\hbar}H|\psi(t)\rangle\,dt - \frac{1}{2}\Gamma\left(c^\dagger c - \langle c+c^\dagger\rangle c + \frac{1}{4}\langle c+c^\dagger\rangle^2\right)|\psi(t)\rangle\,dt + \sqrt{\Gamma}\left(c - \frac{\langle c+c^\dagger\rangle}{2}\right)|\psi(t)\rangle\,dW \tag{88}$$

Because our discrete collisional model extracts a single real bit of information per collision (projecting the ancilla into a specific basis), its continuous limit intrinsically falls into this category, generating a real stochastic variable.

Heterodyne Detection (Complex Noise Unraveling)

If the measurement extracts information on two conjugate quadratures simultaneously (heterodyne detection, corresponding to the original Gisin-Percival QSD formulation), the unraveling is driven by a complex Wiener increment $d\xi = (dW_1 + idW_2)/\sqrt{2}$. This complex noise satisfies $|d\xi|^2 = dt$ and $d\xi^2 = 0$. In this case, preserving the norm requires balancing the full complex expectation value $\langle c\rangle$:

$$d\,|\psi(t)\rangle = -\frac{i}{\hbar}H\,|\psi(t)\rangle\,dt - \frac{1}{2}\Gamma\left(c^\dagger c - 2\langle c^\dagger\rangle c + \langle c^\dagger\rangle\langle c\rangle\right)|\psi(t)\rangle\,dt + \sqrt{\Gamma}(c - \langle c\rangle)\,|\psi(t)\rangle\,d\xi \tag{89}$$

In both continuous formulations, the non-linear terms containing the expectation values act as a continuous feedback. They dynamically compensate for the non-unitary drift generated by the measurement operator c, strictly preserving the normalization of the wave function at every infinitesimal time step dt.

5.3 Linear (Unnormalized) Stochastic Equations

In analytical calculations and numerical simulations, it is often much more efficient to use the linear (or homogeneous) versions of the SSEs. In this approach, we drop the non-linear expectation value terms, leading to the evolution of an unnormalized state vector $|\tilde{\psi}(t)\rangle$. The physical state is then recovered by manually normalizing the vector: $|\psi(t)\rangle = |\tilde{\psi}(t)\rangle\,/\sqrt{\langle\tilde{\psi}(t)|\tilde{\psi}(t)\rangle}$.

To write these equations compactly, we introduce the non-Hermitian effective Hamiltonian:

$$H_{\text{eff}} = H - i\frac{\hbar}{2}\Gamma c^\dagger c \tag{90}$$

For the **Quantum Jump** unraveling, the linear SSE simplifies drastically. The state evolves under the effective Hamiltonian until a jump occurs, without any continuous norm-preserving feedback:

$$d\,|\tilde{\psi}(t)\rangle = \left[-\frac{i}{\hbar}H_{\text{eff}}dt + (c - \mathbb{I})\,dn\right]|\tilde{\psi}(t)\rangle \tag{91}$$

Similarly, for the **Quantum State Diffusion** driven by real noise $d\xi$, the removal of the continuous feedback terms $\langle c\rangle$ and $\langle c^\dagger c\rangle$ yields a purely linear stochastic differential equation:

$$d\,|\tilde{\psi}(t)\rangle = \left[-\frac{i}{\hbar}H_{\text{eff}}dt + \sqrt{\Gamma}c\,dW\right]|\tilde{\psi}(t)\rangle \tag{92}$$

These linear forms clearly show that the deterministic drift between measurements is universally governed by the same non-Hermitian effective Hamiltonian, regardless of the chosen unraveling.

References

[1] E. Coccia, F. Troiani, and S. Corni, "Probing quantum coherence in ultrafast molecular processes: An *ab initio* approach to open quantum systems", The Journal of Chemical Physics **148**, 204112 (2018).

[2] K. Mølmer, Y. Castin, and J. Dalibard, "Monte Carlo wave-function method in quantum optics", Journal of the Optical Society of America B **10**, 524 (1993).